

Research interests: Social justice applications of data science, reduced-order modeling, data-driven modeling, machine learning, modal decompositions, transport-dominated physics, traveling waves, fluid dynamics

Education

University of Washington

2016-2021

Ph.D. in Mechanical Engineering, M.S. 2018

Seattle, WA

Thesis: Data-Driven Reduced-Order Modeling for Symmetrical Systems

Advisors: Steven Brunton (Mechanical Engineering) and J. Nathan Kutz (Applied Mathematics)

Seattle University

2012-2016

B.S. in Mechanical Engineering, Minor in Mathematics

Seattle, WA

Senior Design project with Ingersoll Rand, resulting in patent

Research advised by John Carter (Mathematics), supported by Clare Boothe Luce Foundation Undergraduate Research Award

Teaching

Instructor

DATA 5310 **Data Visualization** F23, F22
DATA 5901 **Capstone I** W26, W25, W24, W23
DATA 5902 **Capstone II** S25, S24, S23
DATA 5322 **Statistical Machine Learning II** S25, S24, S23, S22
DATA 3310 **Data Visualization** W26, W24, F22
MATH 1335 **Calculus II** W22, F21

Capstone Advisor

MSDS 25.2 **Janani Krishnamurthy, Satwik Kunaparaju, Ting-Yu Lin, Prasanna Nalabothu, Surya Ramesh**
Planning and Mission Effectiveness Demographic Data History, Archdiocese of Seattle
MSDS 25.5 **Lavanya Bunadri, Garrett Ringler, Ruqhaiya Syeda, Maelice Yamdjieu**
A Modern Approach to Pacific Salmon Research: An Online Dashboard, NOAA Fisheries
MSDS 24.3 **Wai Chun Chong, Malia Cortez, Tuan Anh Nguyen, Anna Tsai**
Meta-Analysis of Elwha River Water Temperature Regime, NOAA Fisheries, Fish Ecology Division
MSDS 24.5 **Yen-Ting Chen, Hanisha Karri, McKenzie Maidl, Justus Ngunjiri**
Washington State Craft Brewing Industry Snapshot, Washington Brewers Guild
MSDS 23.1 **Liya LaPierre, Karthika Selvaraj, Aishwarya Saibewar, and Radhika Magaji**
Donor Database Analysis and Recommendations, Camp Korey
MSDS 23.3 **Prateek Kakkar, Ramgopal Reddy Putta, and Xuanzhi Li**
A Data Curation System and Predictive Model for Unconfined Concrete Strength, DeSimone Consulting Engineers

Publications

In Progress

Community-Driven Data Science Practices

La Matematica

A Barreda II, C Diaz Eaton, S Hansen, JE Hibdon, L Gordon, R Greenwald, M José Gutiérrez Paz, K İnce, C Kelling, A

Accepted 2026

Mendible, J Mercado, Victor Piercey, Bianca Thompson

Evaluating Remote Sensing and Machine Learning Methods for Predicting Understory Biomass in Post-Harvest Forests

SE Dixon, RJ McGaughey, **A Mendible**, BT Bormann, C Bobsin and GJ Ettl

Paper in preparation

Complete Data for Studying the Role of Demographics in Selection of the Nobel Prizes

A Mendible, J Higdon, M Nagy, R Roca, N Rodriguez, L Xian, C Topaz

Paper in preparation

Peer-Reviewed Journal Publications

Data collection and analysis for small-town policing: challenges and recommendations

Statistics and Public Policy

C Kelling, A Haensch, **A Mendible**, S Brooks, A Wiedemann, M Aminian, W Hasty, J Higdon

2024

DOI: 10.1080/2330443X.2024.2379270

Data science and social justice in the mathematics community

Notices of the Am Math Soc

Q Jones, AR Vindas Meléndes, **A Mendible**, M Aminian, HZ Brooks, N Alexander, C Diaz Eaton, P Chodrow

2023

DOI:10.1090/noti2773

Data-driven modeling of two-dimensional detonation wave fronts

Wave Motion

A Mendible, W Lowrie, SL Brunton, JN Kutz

2022

DOI: 10.1016/j.wavemoti.2021.102879

Data-driven modeling of rotating detonation waves

Phys Rev Fluids

A Mendible, J Koch, H Lange, SL Brunton, JN Kutz

2021

DOI: 10.1103/PhysRevFluids.6.050507

Dimensionality reduction and reduced-order modeling for traveling wave physics

Theor Comput Fluid Dyn

A Mendible, SL Brunton, AY Aravkin, W Lowrie, JN Kutz

2020

DOI: 10.1007/s00162-020-00529-9

Reduced strain mechanochemical activation onset in microstructured materials

Polym Chem

JJ Schwartz, R Behrou, B Cao, M Bassford, **A Mendible**, C Shaeffer, AJ Boydston, N Boechler

2020

DOI: 10.1039/C9PY01875A

Randomized nonnegative matrix factorization

Pattern Recognit Lett

NB Erichson, **A Mendible**, S Wihlbom, JN Kutz

2018

DOI: 10.1016/j.patrec.2018.01.007

Professional Articles

Connecting the Dots: A Preliminary Network Analysis of Complaints about Police in Minneapolis

DSPACE Blog

C Kelling, **A Mendible**, M Wang, A Wiedemann, T Marrinan, E Pederson

2025

Counting on Justice: Bridging Quantitative Research and Community Empowerment

MAA Focus

A Mendible, C Kelling

2024

Quantitative Justice: Using Data Science for Good

MAA Math Values

A Mendible

2024

Getting Away From One Right Answer

COMAP Consortium Newsletter

A Mendible

2022

Patents

Air Flow Hour Meter

US Patent 10,713,858

A Mendible, S Heard, T Tran, A Hardesty

2017

Presentations

Quantitative Justice Through R: Data Provenance

Upcoming 2026

AMS Engaged Pedagogy Series

Virtual

Data Science Tools for Community-Driven Police Accountability

Jan 2025

NAM-SIAM-AMS Special Session on Quantitative Justice, Joint Mathematical Meetings

Seattle, WA

Data Science Tools for Community-Driven Police Accountability

Jul 2024

Quantitative Justice through Applied Data Science Minisymposium, SIAM Annual Meeting

Spokane, WA

Panel: Decoding Ethics: Perspectives on Responsible Data Science

Women In Data Science Puget Sound

May 2024

Seattle, WA

Small Town Police Accountability: Case Studies and Opportunities in Data Science

Jan 2024

NAM-SIAM-AMS Special Session on Quantitative Justice, Joint Mathematical Meetings

San Francisco, CA

Data Science for Justice: Small Town Policing

Nov 2023

QSIDE Datathon4Justice, Opening Plenary

Virtual

Small Town Police Accountability: A Data Science Toolkit

Jul 2023

SciPy: Scientific Computing with Python

Austin, TX

Using Data Science for Social Justice

Mar 2023

Annual Meeting of the PNW Section of the Mathematical Association of America

Newberg, OR

QSIDE's STOPA (Small Town Policing Accountability) Research Lab

Oct 2022

QSIDE Datathon4Justice, Opening Plenary

Virtual

Meet Your Data Science Faculty

Oct 2022

Seattle University Mathematics Seminar

Seattle, WA

The Role of QSIDE's STOPA Research Lab in the Field of Criminal Justice

Apr 2022

QSIDE Data4Justice Conference

Virtual

Reduced-Order Modeling

Mar 2022

Pacific Lutheran University Mathematics Seminar

Virtual

Data-driven Modeling of Shock Waves in Rotating Detonation Engines

Mar 2021

SIAM Computational Science and Engineering

Virtual

Data-driven Modeling of Traveling Waves

Dec 2020

University of Delaware SIAM Seminar Series

Virtual

Data-driven Modeling of Traveling Waves

2020

Steve Brunton channel video abstract (14k views)

YouTube

Data-driven Modeling of Detonation Wave Interactions in Rotating Detonation Engines

Nov 2020

73rd Annual Meeting of the APS Division of Fluid Dynamics

Virtual

Dimensionality Reduction and Reduced Order Modeling for Traveling Wave Physics

Nov 2019

72nd Annual Meeting of the APS Division of Fluid Dynamics

Seattle, WA

Poster: The Space-Time Problem for Model Reduction of Traveling Waves

Jun 2019

DTRA Technical Program Review

Fairfax, VA

The Space-time Problem with Model Reduction for Traveling Waves

Mar 2019

SIAM Computational Science and Engineering

Spokane, WA

Scholarly Development

Grant Funding

Optimization of Cryoprotective Agents for Single-Mode Electromagnetic Resonant (SMER)

\$50,000

Rewarming Technology to Enable Long-Term Organ Preservation

Seattle U College of Science and Engineering

2023

Multidisciplinary Research Program Award with Shen Ren (Mechanical Engineering)

Small Town Police Accountability Lab

Workshop hosting

Banff International Research Station for Mathematical Innovation and Discovery

2022

Proposal accepted

Workshops

Reproducible Machine Learning Workflows for Scientists

Aug 2025

US Research Software Sustainability Institute, Funded Participant

Madison, WI

Community Engaged Learning Faculty Fellowship Workshop

Sep 2023

Seattle U Center for Community Engagement, Funded Participant

Seattle, WA

Educating at the Intersection of Data Science and Social Justice Workshop

Jul 2023

Institute for Computational and Experimental Research in Mathematics, Brown University, Funded Participant

Providence, RI

Community Engaged Learning Workshop Series

May 2023

Seattle U Center for Community Engagement

Seattle, WA

Data Science and Social Justice: Networks, Policy, and Education Workshop	<i>Jun-Jul 2022</i>
Institute for Computational and Experimental Research in Mathematics, Brown University, Funded Participant	<i>Providence, RI</i>
Project NExT Fellow	<i>2021-2022</i>
Mathematical Association of America	<i>Virtual</i>
Preparing for Careers in Teaching Statistics and Data Science Workshop	<i>Aug 2021</i>
Section on Statistics and Data Science Education of the American Statistical Association	<i>Virtual</i>
EPSRC Summer School: Modal Decompositions in Fluid Mechanics	<i>Aug 2019</i>
Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Funded Participant	<i>Cambridge, UK</i>

Research Students

Jesse Loi	<i>AY25-26</i>
Data Accessibility for Public Accountability, supported by SU Technology Ethics Initiative Student-Scholar Program	<i>MS Data Science</i>
David Stanko and Mark Daza	<i>AY25-26</i>
Applications of Language Models for Complaint Data: NER and LLMs	<i>MS Data Science</i>
Emma Oriol	<i>AY23-24</i>
Redlining and Gentrification	<i>MS Data Science</i>
Alexander Churhill	<i>AY23-24</i>
Historical Redlining and Housing Quality in Seattle	<i>MS Data Science</i>
Chang He	<i>AY23-24</i>
Historical Redlining and Cycling Accessibility	<i>MS Data Science</i>
Malia Cortez	<i>AY23-24</i>
Historical Redlining and Population Flows	<i>MS Data Science</i>

Service

Academic Service

Semester Transition Coordinator, Data Science	<i>2025-present</i>
Lead transition from quarters to semesters for data science programs	
MS Data Science Online Course Creation	<i>2025-present</i>
Create material for four asynchronous courses for online MSDS program	
MS Data Science Student Admissions Review	<i>2022-present</i>
Review ~250 student applications per year, recommended admissions decisions	
Chile Study Abroad Trip Co-Lead	<i>2024, 2026</i>
Assisted in supervising spring break student trip for Mathematical Models of Near-Shore Phenomena course	
Bannan Building Safety Committee	<i>2025-present</i>
Math Department Website Review	<i>2025</i>
Math Department Community Building	
Organize end-of-year student celebration for graduating MSDS cohort	<i>2023, 2024</i>
Organize weekly (AY 22-23) or quarterly (AY 23-24) activities and events to build departmental community	<i>2022-2024</i>
Data Science Faculty Hiring Committee	
Hosted 6 all-day candidates, 2 hires	<i>2025</i>
Reviewed 54 candidate applications, 14 screening interviews, hosted 4 all-day candidates, 2 hires	<i>2023</i>
Reviewed ~9 candidate applications, 9 screening interviews, hosted 2 all-day candidates, position unfilled	<i>2022</i>
Math Department Recruitment, Inclusion, Diversity, Equity and Retention Committee	<i>2021-present</i>
Member, led climate survey creation and distribution	
QSIDE Institute Consortium SU Liaison	<i>2021-present</i>
Lead event coordination, advertisement at SU, and student involvement	

Professional Service

QSIDE Data Science, Police Accountability and Community Engagement Research Lab	<i>2021- present</i>
Co-Director, lead meetings and organize research activities	
SciPy Conference	

Program Committee, organize topics and oversee tracks conference-wide	2025, 2026
Track Chair: Human Networks, Social Sciences, and Economics	2024
Women In Data Science Puget Sound	
Invited mentor for Mentorship Night for career-seeking women	2024
QSIDE Datathon4Justice	
Co-Organizer	2023
Team Lead	2021, 2022, 2023
La Matematica	-
Referee	
Humanities and Social Sciences Communications	-
Referee	
Journal of Fluid Mechanics	-
Referee	
Community Service	
Mechanical Engineering Students Against Racism , UW Mechanical Engineering	2020-2021
Crafters Against CoVID-19	2020
Engineering Discovery Days , UW College of Engineering	2016, 2019
Bike & Build Drift West	2017
UW Society of Women Engineers , Elementary school outreach visits	2016

Honors & Awards

Community Engaged Learning Faculty Fellowship , Seattle U Center for Community Engagement	AY 23-24
Supervisor of the Year , Seattle University Student Employment Shine Award	2023
Project NExT Fellow , Mathematical Association of America	2021
Graduate Diversity Fellowship , UW Graduate Opportunities and Minority Achievement Program, Full Support	2016, 2020
Steve and Lynn Pratt Fellowship , UW Mechanical Engineering	2017, 2020
Wynne Alexander Guy Spirit of the Mathematics Department Award , SU Mathematics	2019
Wei Li and Dongmei Chen Endowed Fellowship , UW Mechanical Engineering	2017
Brian and Mary Fabien Endowed Fellowship , UW Mechanical Engineering	2016, 2017
Undergraduate Research Award , Clare Boothe Luce Foundation	2014

Additional Experience

Clear Motion (formerly Levant Power)	June - Aug 2016
Controls Engineering Intern	Woburn, MA
National Renewable Energy Laboratory	June - Aug 2015
Science Undergraduate Laboratory Internship	Boulder, CO
Network for Earthquake Engineering Simulation	June - Aug 2014
OH Hinsdale Wave Research Laboratory REU	Corvallis, OR